
asianpaints
ALCHEMY
Platform Partner unstop

WHITER WITH LESS TiO2

Case Study 1: Future of Whiteness Development through Material Science

OUR ANSWER: KEEP THE RUTILE, HOLD IT IN A NETWORK OF AIR, AND SEAL THE AIR IN WITH THE SECOND COAT.

Rutile scatters 1.52x more light per unit volume in air than in binder. A porous first coat holds 9.1% TiO2 in 35% air; a 16.4 µm seal coat keeps the air in and gives the washable surface. Our optical model matches a premium white's contrast ratio and brightness with 24.8 g of TiO2 per m² instead of 40.8 g.

TEAM IDEAFORGE

Team Members	Institution	Role
Harsh Kawatra	Delhi Technological University (DTU)	Team Leader
Anuj Gambhir	Delhi Technological University (DTU)	Member

Read the full research document →

WHO SHAPES A WHITE WALL

Asian Paints buys about 1,19,919 t of TiO2 a year, our estimate, and 57% of India's supply came from China in DGTR's probe.

Logos identify organisations in the analysis; no partnership is implied.

PIGMENT
Index 2.75
rutile bends light most; binder is 1.49

OPACIFIERS
9% less TiO2
from hollow polymer spheres, in our screen

RIVALS
Over 21%
of industry capacity is now Birla Opus

SCIENCE
5 to 15 µm
beetle scales, white with no pigment

asianpaints
40.8 g TiO2
a m² today

TRADE RULES
$460 to $681
a tonne duty, asked again on Chinese TiO2

PAINTERS
13.42 m²
a litre a coat; same rollers as today

HOMEOWNERS
78% repaint
of paint demand goes over an old wall

STANDARDS
IS 15489:2024
the BIS spec for emulsion paint

Source: Alchemy Case Study 1 | model b377745d37 | Burresi et al. 2014 | Vukusic et al. 2007 | DGTR 3 Aug 2026 | IS 15489:2024 | Apcolite TDS | photos and logos: Wikimedia Commons, credits in research Appendix D.12

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THE TECHNOLOGY

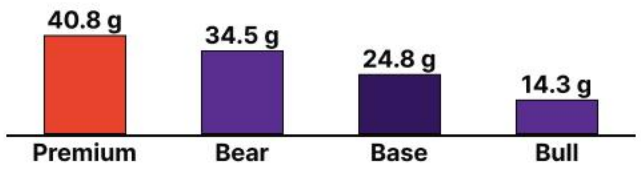
Keep rutile and hold it in sealed air: a two-coat white that matches a premium finish with 39% less TiO2 and 34% lower material cost per m²

OUR ANSWER Keep the rutile, move it into a network of air and seal the air in, inside the two coats painters already apply.

	THE BRIEF ASKS	THE OBVIOUS MOVE	WHAT WE RECOMMEND	THE PROOF	THE RESULT
1	**MAKE IT WHITER** Opacity, challenge W-K1 Slides 2 to 4	~~Find a new pigment~~ 0 of 5 TiO2-free designs reach Y 90.7; zinc sulfide, the closest, gets 90.4.	**Move the TiO2 into air** A base coat above its critical PVC is 35% air, so rutile sees index 1.18, not 1.49.	Scattering per µm³ of rutile Binder, 1.49 — 8.2 Ours, 1.18 — 10.9 Air, 1.00 — 12.6	**39%** less TiO2 per m² at Y 90.7 and CR 0.998, in our model
2	**CUT THE BILL** Supply, economics, W-K4 Slides 4 and 7	~~Wait for cheap TiO2~~ 57% of Indian demand came from China; a duty of $460 to $681 a tonne is recommended again.	**Buy less of it** 16.1 g less TiO2 and 17.5 g less binder per m²; finer chalk adds back ₹1.2.	Material cost per m², two coats, ₹ Premium — ₹18.4 Ours — ₹12.0 34% lower, 16% to 51% across cases	**₹298 Cr** a year at 40% adoption; 7,551 t of TiO2 not bought, estimate
3	**MAKE IT AT SCALE** Manufacturing, W-K5; W-Q4 to Q7 Slides 6 and 7	~~Build a new particle~~ Yolk-shell and hollow particles need a new synthesis line and save at most 20% of TiO2.	**Same plant, same coats** Base coat and seal coat run on today's dispersers; painters apply two coats as now.	TiO2 saved under the same test Hollow polymer — 9% Nano-air binder — 11% Hollow rutile — 20% Ours — 39%	**12 months** to an interior pilot; lab panels and Gate 0 by month 6

IN TOTAL

TiO2 PER m², TWO COATS, g



THREE CASES, ONE ANSWER

Case	Bear	Base	Bull
TiO2 saved	15%	39%	65%
Material cost cut	16%	34%	51%
Net saving a year	₹38 Cr	₹298 Cr	₹1,021 Cr
TiO2 avoided a year	928 t	7,551 t	25,934 t

WHY CASE STUDY 1

1 Whiteness — 4.50
2 Thermal — 3.30
3 Bio-based — 3.25
4 AI, data — 3.05

Weighted, team judgement

THE FIRST STEP

By month 6: 27 panels in a 3 by 3 test of porosity and seal infiltration, read for Y 90.7 and CR 0.998 at Gate 0.

Then a pilot batch on existing lines.

Source: team optical model b377745d37 (T4, T8, T9, T12, T13) | DGTR findings 2025, 2026 | case scores are team judgement, research section 8 | research sections 2, 8 and 12 to 23

THE CHALLENGE

Rutile in binder scatters only 66% of the light it would in air, and crowding at 18% costs another 9%: the binder, not the pigment, caps whiteness

THE BILL

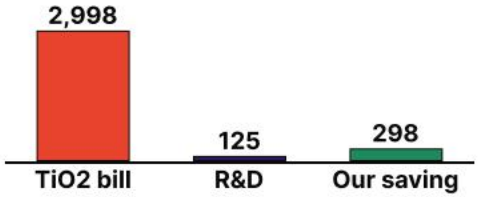
WHAT TiO2 COSTS ASIAN PAINTS

₹2,998 Cr a year at 15% of raw materials, estimate

1,19,919 t of TiO2 a year at ₹250 a kg, estimate

40.8 g of TiO2 in each m² of premium white

₹ CR A YEAR, AGAINST R&D



THE DUTY CLOCK ON CHINESE TiO2



So what: 57% of Indian demand came from China.

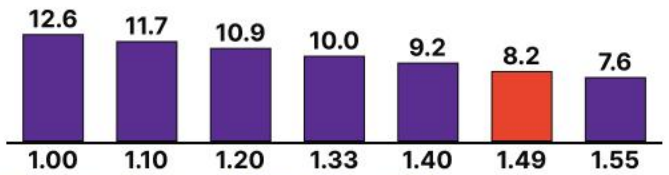
DUTY ASKED, $ A TONNE



Recommended, not yet in force

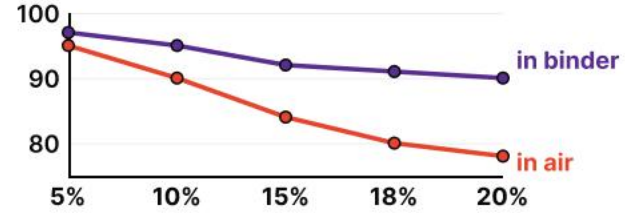
THE PHYSICS

THE MEDIUM SETS THE SCATTERING



Per µm³ of rutile: air 1.00, binder 1.49

CROWDING COSTS EFFICIENCY, %



FIVE FORCES ON A PAINT MAKER'S TiO2

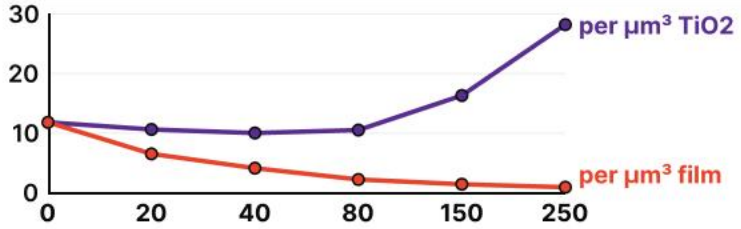
Supplier power	High	57% from China; duty in dispute
New entrants	High	Birla Opus: over 21% of capacity
Rivalry	High	JSW to buy up to 74.76% of Akzo India
Buyer power	Medium	Repaint is 78% of decorative demand
Substitutes	Low	0 of 5 TiO2-free whites reach Y 90.7

THE TRAP

THE USUAL FIXES, UNDER ONE TEST

Fix	Saved	Why it stops
Hollow polymer	9%	air sits in a shell
Nano-air binder	11%	index barely falls
Hollow rutile	20%	walls too thin
Yolk-shell TiO2	none	misses Y: 86.1
High BaSO4	none	1.64, near binder

AIR GAP: TiO2 GAINS, THE FILM LOSES



WHERE SHOULD THE AIR LIVE

Inside each particle: its own pocket costs 9x the film

In the binder: closed nano-air saves only 11% of TiO2

In the film: one shared air network costs nothing

So the air has to live in the film, not in the particle.

Source: team optical model b377745d37 (T4 to T8) | Asian Paints FY26 | DGTR Feb 2025, Aug 2026 | Calcutta HC Sep 2025 | CBIC 33/2025 | estimates A13, A21 to A23 | force ratings are team judgement

WHY IT IS UNIQUE

We ran 12 white-film designs through one model: no TiO2-free design reaches premium brightness, and only an air network under a seal coat saves more than 30% of TiO2

THE SCREEN

TWELVE DESIGNS, ONE TEST, LEAST TiO2 WINS

Design	Family	TiO2 a m²	Saved	Y	CR	Verdict
1 Rutile in binder (premium today)	Reference	40.8 g	0%	90.7	0.998	Reference
2 Rutile plus hollow polymer spheres	Commercial	37.1 g	9%	90.8	0.998	Drop
3 Rutile in a nano-air binder	Commercial	36.3 g	11%	90.7	0.998	Drop
4 Rutile in its own air pocket	Particle	misses	n/a	86.1	0.983	Drop
5 Hollow rutile spheres	Particle	32.9 g	20%	90.7	0.998	Drop
6 Zinc sulfide, no TiO2	Substitute	misses	n/a	90.4	0.997	Drop
7 Zirconia, no TiO2	Substitute	misses	n/a	88.0	0.993	Drop
8 Barium sulfate, high loading	Substitute	misses	n/a	63.2	0.489	Drop
9 Organic crystals (shrimp-eye type)	Substitute	misses	n/a	84.1	0.975	Drop
10 Air network under a seal coat (ours)	Film	24.8 g	39%	90.7	0.998	Keep
11 Air network, no TiO2 at all	Film	misses	n/a	82.9	0.973	Drop
12 Air network with organic crystals	Film	7.9 g	81%	90.7	0.998	Later

Reference: 1 | Commercial 2: 0 kept | Particle 2: 0 kept | Substitute 4: 0 kept | Film 3: 2 kept

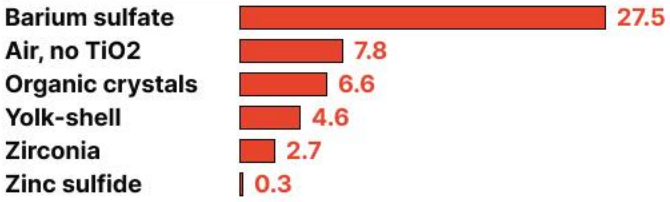
Design 12 saves more (81%) but uses an unverified pigment, kept for horizon 3, not this recommendation.

THE TEST AND THE MODEL

Pass: Y 90.7 or more and CR 0.998 or more, two 30 µm coats over black and white; the least TiO2 wins.

- Mie: each particle
- Percus-Yevick: crowding
- Kubelka-Munk: two coats
- CIE D65: colour and CR

HOW FAR EACH MISS FALLS SHORT, Y



BaSO4, the brief's own example, misses by the most.

THE VERDICT

FIVE CHALLENGES, THREE ROUTES

Route for	Pigment	In particle	In film
Opacity	No	Part	Yes
Durability, yellowing	Part	Part	Test
Dispersion	No	Part	Yes
Supply, economics	No	Part	Yes
Manufacturing	No	No	Yes

PRIOR ART: WHO HAS WHICH PIECE

Prior art	Air	Seal	TiO2	Rule
Hollow polymer spheres	Part	No	Yes	No
Spacer-coated TiO2, Dow	No	No	Yes	No
Above-CPVC extenders	Yes	No	Yes	No
Porous cooling paint	Yes	No	No	No
Multilayer, TiO2-free	Part	Yes	No	No
This design	Yes	Yes	Yes	Yes

THE TAKEAWAY

No new pigment out-scatters rutile in binder: zinc sulfide, the closest, misses brightness (90.4 against 90.7). Air inside each particle costs too much film. The gain is in where the film keeps its air.

"biologically benign replacements for artificial scatterers like TiO2" Lemcoff et al., Nature Photonics, 2023

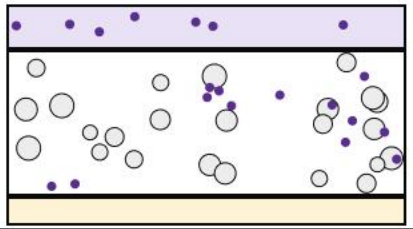
Source: team optical model b377745d37 (T8, same targets for all 12) | prior art: US5147940A, Dow EVOQUE, Imerys Celtix, Mandal et al. 2018, US9636706B2 | Lemcoff et al. 2023 | research sections 11 to 13

MECHANISM AND PROOF

A porous first coat holds 9.1% TiO2 in 35% air under a 16.4 µm seal coat, and matches the premium white with 24.8 g of TiO2 per m² against 40.8 g

HOW IT WORKS

THE FILM, IN SECTION



1. Seal 16.4 µm: latex bigger than pores
2. Base 46.2 µm: 35% air, rutile, chalk
3. Rutile sees index 1.18, not 1.49
4. Same wet film; air is free volume

WHAT THE RUTILE SEES

Case	Air	Index	Gain	Saved
Bear	25%	1.38	1.12x	15%
Base	35%	1.18	1.33x	39%
Bull	45%	1.00	1.52x	65%

Gain: scattering against rutile in binder



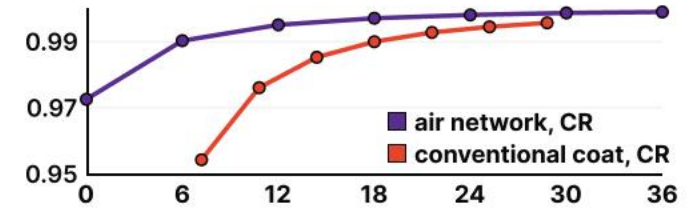
NATURE GOT HERE FIRST

The Cyphochilus beetle's scales: chitin filaments in air, about 5 µm thick, white with no pigment. Our base coat copies the network; the seal coat closes it.

"the beetle scales exhibit the lowest transport mean free path length" Burresi et al., 2014

THE PROOF

6 g IN AIR HIDES LIKE 18 g IN BINDER



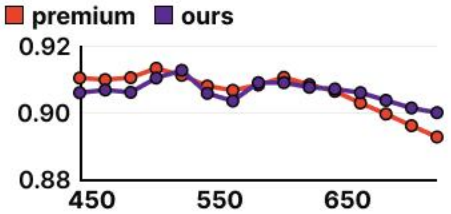
TWO COATS, THE SAME WHITE

39% LESS TiO2

40.8 g to 24.8 g a m² at CR 0.998 and Y 90.7

Bear 15% | Base 39% | Bull 65%

THE SAME SPECTRUM



AGAINST THE TARGETS

Measure	Premium	Ours
Contrast ratio	0.9984	0.9980
Brightness Y	90.7	90.7
L*	96.3	96.3
b*, yellowness	1.4	1.5
TiO2 a m²	40.8 g	24.8 g
NIR, 1,500 nm	63.7%	71.1%

HOW IT IS MADE

FOUR STEPS ON EQUIPMENT ASIAN PAINTS ALREADY RUNS

1 FINE CHALK	2 BASE COAT	3 SEAL COAT	4 ON THE WALL
Indian PCC makers 0.6 µm precipitated chalk from lime and CO2	plant dispersers TiO2 and chalk above critical PVC, binder 25% of solids	letdown tanks acrylic latex larger than the pores, plus 12% TiO2	painters, as today coat 1, let it dry, coat 2; same rollers and rate

THE LAB PLAN, NOT YET RUN: ONE TEST PER CHALLENGE

Brief challenge	Test	Pass mark
Opacity	ISO 6504-3, contrast ratio	Y 90.7 and CR 0.998 or more
Durability, yellowing	QUV weathering, ASTM D2486	b* 1.4 ±0.2; scrub as premium
Dispersion	porosimetry after storage	porosity stays 25 to 45%
Supply, economics	pilot batch costing	materials near ₹12.0 a m²
Manufacturing	pilot on existing lines	no new unit operation

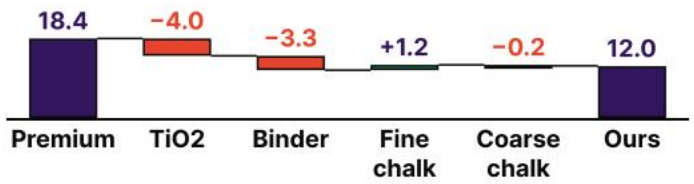
Source: team optical model b377745d37 (T7, T9 to T11, T15) | Burresi et al. 2014, Sci. Rep. 4, 6075 | ISO 6504-3, ASTM D2486 | research sections 14 to 18 and 22 | photo: A. Parnell, Wikimedia Commons, CC BY-SA 4.0

COST AND SCALE

At 40% adoption in interior emulsions the system saves about ₹298 Cr a year in materials and 7,551 t of TiO2, with no new plant or painter training

COST

MATERIAL COST PER m², TWO COATS, ₹



₹18.36 to ₹12.04 before rounding, to the nearest ₹0.1.

RAW MATERIALS AND HANDLING

Material	What changes	Handling
Rutile TiO2	less of it	same dust controls
Fine chalk (PCC)	lime plus CO2	not GHS-classified
Acrylic latex	larger latex	closed storage, PPE
Silane, exterior	keeps pores dry	diluted in water

TiO2 limits: OSHA 15 mg/m³; NIOSH 2.4 mg/m³ fine

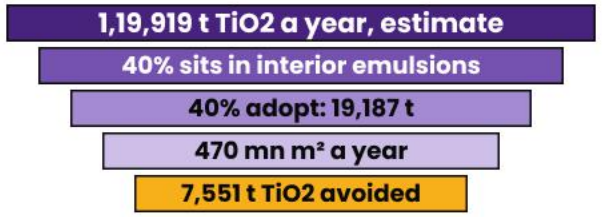
THREE SCENARIOS

Case	TiO2 cut	₹ a m²	₹ a year	TiO2 a year
Bear	15%	₹13.7	₹38 Cr	928 t
Base	39%	₹12.0	₹298 Cr	7,551 t
Bull	65%	₹10.0	₹1,021 Cr	25,934 t

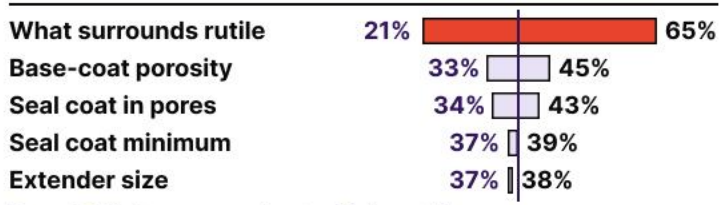
Per litre, two coats: ₹123.5 to ₹80.5 at 13.42 m² a litre a coat.

SCALE

FROM TiO2 BOUGHT TO TiO2 AVOIDED



WHAT MOVES THE SAVING MOST, % SAVED



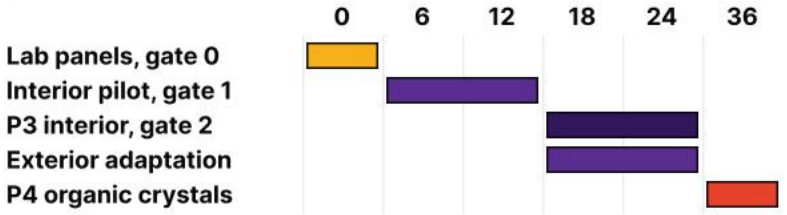
Base 39%; bars span low to high setting

WHERE IT APPLIES, AND HOW READY

Segment	Medium	Phase	Ready
Interior, white, new walls	water	Phase 2	Highest
Repaint, 78% of demand	water	Phase 3	Medium
Interior, tinted	water	Phase 3	Medium
Exterior emulsions	water	Phase 3	Lower
Industrial, automotive	solvent	Phase 4	Lowest

ROLL-OUT

THREE HORIZONS, THREE GATES, BY MONTH



12 RISKS, IMPACT BY PROBABILITY

	Low	2	Med	4	High
High			1 2 4 10		
4					
Med		7 11	5 6 8	3	
2			9		
Low	12				

TOP RISKS, EARLY SIGNAL, RESPONSE

Risk	Early signal	Response
Binder wets rutile	FIB-SEM at gate 0	leaner base binder
Seal fills pores	interface FIB-SEM	latex 2x pore size
Thin seal coat	film below 15 µm	enforce 15 µm floor

Therefore: panels in 6 months, an interior pilot by month 12; no new pigment, plant or painter skill.

Source: team model b377745d37 (T12 to T14) | Asian Paints FY26 | Apcolite TDS coverage | TiO2 price, share and adoption are team assumptions A13, A23 to A25 | risk register, research section 24

ANNEXURE

Annexure: one optical model behind every number; the answer holds from 15% to 65% of TiO2 saved

A1 MODEL CHAIN AND CHECKS

Mie, multilayer	particles
Percus-Yevick	crowding
Kubelka-Munk	two coats
CIE D65, 2 degree	colour
Optimiser	least TiO2

- **Mie:** Bohren and Huffman benchmark within 0.1%
- **Layers:** two-layer sphere equals a solid one
- **Crowding:** Percus-Yevick limit reproduced
- **Films:** Kubelka-Munk thick-film limit exact
- **Run:** hash b377745d37, 13 of 13 pass

A2 ASSUMPTIONS, BEAR TO BULL

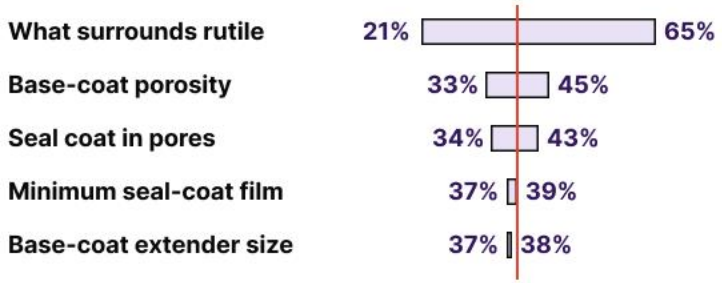
ID and assumption	Bear	Base	Bull
A05 Base-coat porosity	25%	35%	45%
A07 Seal coat in the pores	25%	10%	0%
A08 What surrounds rutile	binder-wetted	air, binder	air
A28 Minimum seal coat	20 µm	15 µm	12 µm
A13 TiO2 price a kg	₹200	₹250	₹300
A23 TiO2 share, materials	10%	15%	20%
A24 Addressable share	30%	40%	50%
A25 Adoption by year 5	20%	40%	60%

Bear is the value least favourable to the recommendation.

A2 WHAT WOULD CHANGE OUR MIND

- **Porosity:** measured outside 25 to 45%
- **Surroundings:** binder wets rutile; saving falls to about a fifth
- **Seal coat:** under 12 to 20 µm fails to form a skin
- **Patent:** our exact combination already claimed
- **Prices:** TiO2 price or share outside bear to bull

A3 SENSITIVITY, NEAR-INFRARED AND ONE-INPUT MOVES



Base 39%; bars span one input, low to high

Wavelength	Premium	Ours
800 nm	87.0%	87.9%
1000 nm	82.8%	86.3%
1200 nm	73.3%	79.4%
1500 nm	63.7%	71.1%

If this moves	₹ Cr a year
Adoption 20%	149 net
Adoption 60%	447 net
TiO2 at ₹200 a kg	151 on TiO2
TiO2 at ₹300 a kg	227 on TiO2

A4 SOURCES AND SHORT FORMS

1. Alchemy Case Study 1 brief and references
2. Bohren and Huffman, 1983 (Mie benchmark)
3. Wertheim, Phys. Rev. Lett. 10, 321, 1963
4. Kubelka and Munk 1931; Saunderson, JOSA 32, 727, 1942
5. Devore, JOSA 41, 416, 1951; Sultanova et al. 2009
6. CIE 15:2018 colorimetry
7. Burresi et al., Sci. Rep. 4, 6075, 2014
8. Lemcoff et al., Nature Photonics, 2023
9. Asian Paints Annual Report FY2025-26
10. DGTR final findings, TiO2 from China, 2025 and 2026
11. Calcutta High Court WPO 148 of 2025; CBIC 33/2025
12. US5147940A, US9636706B2, EP0538411
13. Team IdeaForge research and model b377745d37

TiO2 titanium dioxide; PVC pigment volume concentration; CPVC critical PVC; PCC precipitated calcium carbonate; CR contrast ratio; Y luminous reflectance; L*, b* CIELAB lightness, yellowness; NIR near-infrared; µm micrometre; DGTR Directorate General of Trade Remedies; FIB-SEM focused-ion-beam microscopy; H1 to H3 horizons

Source: model tables T1 to T16 | research Appendix A bibliography and Appendix B tables | model b377745d37

THANK YOU

Whiter with less TiO2: keep the rutile, hold it in air, seal the air in

39%

less TiO2 per m^2 at contrast ratio 0.998 and brightness 90.7, in our model

₹298 Cr

a year in materials at 40% adoption, with 7,551 t of TiO2 not bought, estimate

13 of 13

model checks pass; a lab plan with pass marks is ready for Gate 0

RESEARCH DOCUMENT harshkawatra11.github.io/ap-alchemy/research.pdf →

TEAM IDEAFORGE

Team Members	Institution	Role
Harsh Kawatra	Delhi Technological University (DTU)	Team Leader
Anuj Gambhir	Delhi Technological University (DTU)	Member